

Practical 8

Complex numbers in R

- In **R**, complex numbers are numbers that have both a **real** and an **imaginary** part. They are represented in the form:

$$z = a + bi$$

where:

- a is the **real part**,
- b is the **imaginary part**, and
- i is the imaginary unit (the square root of -1).

They are essential in **engineering, physics, signal processing, and mathematics** when real numbers alone are insufficient (e.g., in wave equations, impedance calculations).

Properties of Complex Numbers

- Let $z = a + bi$, $w = c + di$ Then:

1. Addition

$$z + w = (a + c) + (b + d)i$$

2. Multiplication

$$z \cdot w = (ac - bd) + (ad + bc)i$$

3. Modulus (Magnitude)

$$|z| = \sqrt{a^2 + b^2}$$

4. Conjugate

$$\text{Conj}(z) = a - bi$$

5. Argument (Phase Angle)

$$\text{Arg}(z) = \tan^{-1}(b/a)$$

```
z1 <- 3 + 4i
```

```
z2 <- 1 - 2i
```

```
z_add <- z1 + z2    # Addition
```

```
z_mul <- z1 * z2    # Multiplication
```

```
z_div <- z1 / z2    # Division
```

```
z_pow <- z1 ^ 2     # Power
```

Common Functions for Complex Numbers

Function	Description
$\text{Re}(z)$	Returns the real part
$\text{Im}(z)$	Returns the imaginary part
$\text{Mod}(z)$	Returns the modulus (magnitude)
$\text{Arg}(z)$	Returns the argument (angle in radians)
$\text{Conj}(z)$	Returns the complex conjugate
$\text{abs}(z)$	Same as $\text{Mod}(z)$

Create Complex Numbers in R

Complex numbers in R can be created in two main ways:

1. Using `complex()` function

```
z1 <- complex(real = 3, imaginary = 4)
```

2. Using `i` or `1i` notation

```
z2 <- 3 + 4i
```

Practical Uses

- **Signal Processing:** Fourier transforms use complex exponentials.
- **Electrical Engineering:** Impedance and phasor analysis.
- **Quantum Physics:** Wave functions are complex-valued.
- **Control Systems:** Poles and zeros may be complex.

Write a R program to create a complex number

Problem Statement: Write a program on R numbers

1. Creating a numeric data type
2. Creating an integer
3. Creating a complex number
4. Converting from one type to another

1. Creating a numeric data type (default in R is numeric - double)

```
numeric_val <- 10.5
```

```
cat("Numeric value:", numeric_val, "\n")
```

```
cat("Type of numeric_val:", typeof(numeric_val),  
"\n\n")
```

- **10.5** is a decimal number, which R treats as a **numeric** type by default (specifically, a **double**).
- `cat()` is used to print the value and its type.
- `typeof()` returns the internal type, which will be "double" for numeric values.

- Numeric value: 10.5
- Type of numeric_val: double

2. Creating an integer (by adding 'L' to indicate integer)

```
integer_val <- 20L
```

```
cat("Integer value:", integer_val, "\n")
```

```
cat("Type of integer_val:", typeof(integer_val),  
"\n\n")
```

- The **L** suffix tells R to treat 20 as an **integer**, not a double.
- R does not automatically treat whole numbers as integers unless the L is used.
- `typeof(integer_val)` returns "integer".

- Integer value: 20
- Type of integer_val: integer

3. Creating a complex number

```
complex_val <- 4 + 5i
```

```
cat("Complex number:", complex_val, "\n")
```

```
cat("Type of complex_val:",  
typeof(complex_val), "\n\n")
```

- In R, a **complex number** is created using **i** to represent the imaginary part.
- $4 + 5i$ means **real part = 4, imaginary part = 5**.
- R stores this as a complex type internally.

- Complex number: $4+5i$
- Type of `complex_val`: `complex`

4. Converting from one type to another

Convert numeric to integer

```
converted_integer <- as.integer(numeric_val)
```

```
cat("Converted numeric to integer:", converted_integer, "\n")
```

```
cat("Type of converted_integer:", typeof(converted_integer), "\n\n")
```

Convert integer to complex

```
converted_complex <- as.complex(integer_val)
```

```
cat("Converted integer to complex:", converted_complex, "\n")
```

```
cat("Type of converted_complex:", typeof(converted_complex),  
"\n\n")
```

Convert complex to numeric (real part only)

```
converted_numeric <- as.numeric(complex_val)
```

```
cat("Converted complex to numeric (real part):", converted_numeric,  
"\n")
```

```
cat("Type of converted_numeric:", typeof(converted_numeric), "\n")
```

A. Convert Numeric to Integer

- `as.integer()` converts a numeric (e.g., 10.5) to an integer, **truncating** the decimal part.
- So 10.5 becomes 10.

B. Convert Integer to Complex

Converts integer 20L to complex: 20+0i.

C. Convert Complex to Numeric (Real Part Only)

- Extracts only the **real part** of the complex number.
- So $4 + 5i$ becomes 4.

Data Types in R:

Type	Example	Suffix/Note
Numeric	10.5	Default for decimals
Integer	20L	Use L to indicate
Complex	4 + 5i	Use i for imaginary
Type Convert	as.*()	Convert between types